



Search titles, descriptions



Course overview

▼ 01a - Introduction
1/1 Completed



01 - Introduction
✓ Completed

▶ 01b - Probability theory review
3/3 Completed

▶ 02 - PGM foundations I
0/1 Completed

03 - PGM foundations II

Course overview

Location: rooms 060a and 060b in building 358

Important dates:

Mini-tests - **March 25, May 6**

Project deadline (report+notebook) - **May 15**

Course plan overview (tentative)

The course is divided into 3 main blocks:

- Foundations of probabilistic graphical models (PGMs) and probabilistic programming
- Probabilistic modelling in various contexts (with practical examples of popular models)
- Bayesian inference (exact and approximate)

Week	Topic	Short description	Date
1	Intro to the course + Prob. review	Conditional probability, joint distribution, Bayes theorem, conditional independence, expectation, prior, likelihood, maximum likelihood estimation, etc.	4 Feb
2	PGM foundations	Probabilistic graphical models (PGMs), factorization, conditional independence and inference	11 Feb
3	PGM foundations II	Bayesian inference, conjugate priors, generative processes	18 Feb
4	Freq. vs Bayesian + Prob. Programming + Mixture models	MLE vs. MAP estimation, Bayesian inference, Probabilistic programming; Introduction to Pyro; Mixture models	25 Feb
5	Regression models	Bavesian linear regression. Poisson regression.	4 Mar